Could this be Paediatric Sepsis? (Acute Sites Only)

NHS
Royal Devon University Healthcare
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Trustwide

Has PEWS/SPOT triggered?
- OR is child feverish or looking sick?
- OR is parent/carer very worried?

N → **Lower risk of sepsis**
1. Clinician review
2. Manage as per diagnosis
3. Regular PEWS
4. Reconsider sepsis if deteriorates
5. Safety net before discharge

Y ↓

Could this be an infection? N → Consider other differential

Y ↓

Any 1 of HIGH RISK criteria?

- Unresponsive to social cues/difficult to rouse or keep awake/ altered mental state/behaviour
- Appears very unwell to health professional
- Weak, high pitched or continuous cry
- Grunting, respiration or apnoeic episodes
- SpO_2 < 90% / increased oxygen requirement
- Severe tachypnoea *(see table)*
- Severe tachycardia *(see table)*
- Bradycardia < 60 bpm
- Non-blanching rash/mottled/ashen/cyanotic
- Temperature < 36°C
- If under 3 months, temperature >38°C
- Lactate >2mmol/l

N → **Any 1 of MODERATE RISK criteria?**
- Not responding normally/ no smile
- Reduced activity, very sleepy
- Parental or carer concern
- Moderate tachypnoea *(see table)*
- SpO_2 <92% or nasal flaring
- Moderate tachycardia *(see table)*
- Capillary refill ≥3 seconds
- Reduced urine output (<1ml/kg/hr)
- Pale or flushed
- Leg pain/cold extremities/very pale
- Immunocompromised
- If 3-6 months temperature >39

N ↑

Y ↓

- Send bloods (if 2 moderate criteria present, consider if 1 present): Blood gas (lactate, glucose, BE), blood cultures, FBC, U&Es, LFTs, CRP, coagulation
- Immediate call to Paeds/EM ST3+ to review & results reviewed within 1 hr.

Is lactate >2 ?

Y → RED FLAG SEPSIS!!

Start sepsis 6 pathway NOW

Y → RED FLAG SEPSIS!!

N ↓

- Reassess as per PEWS/SPOT escalation policy (at least hourly)
- Clinician to make antibiotic decision within 3 hours
- Discharge at discretion of senior - with appropriate safety netting advice

Age	Tachypnoea		Tachycardia	
	Severe	*Moderate*	*Severe*	*Moderate*
0-3m	>90	70-89	≥180	160-179
4-11m	≥80	60-79	≥180	170-179
1-4 y	≥70	65-69	≥170	150-169
5-11 y	≥50	40-49	≥150	130-149
12+ y	≥35	30-34	≥130	120-129

Action (complete ALL within 1 hour)

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1 **Give 100% (15L) oxygen via a non-rebreathe mask** (*Unless contraindicated)

Call for early Senior help (ST3+).

If flowsheet is not followed ST3+/Cons to clearly document rationale & LOW threshold for starting flowsheet.

2 **Obtain IV/IO access, take bloods**

- Blood cultures, blood glucose, blood gas (including lactate, glucose, BE), FBC, U&Es **When reviewing blood gas results always assume a lactate >2mmol/l is correct.** Repeat rapidly to confirm the trend.

3 **Give IV/IO/IM antibiotics**

- High Risk: Give immediately, Medium Risk: Always <3 hrs
- According to Trust guidance or national guidance
- Check allergies prior to administration

4 **Consider IV/IO fluids**

- If hypotensive/ lactate > 2mmol/l, give a 10ml/kg bolus and re-assess
- If lactate >4mmol/l also call ICU & WATCh
- Beware of fluid overload! Examine for hepatomegaly, crepitations, gallop, *or respiratory insufficiency.* Be cautious in groups at risk of overload (e.g. neonates and cardiac)

5 **Ensure senior ED/Paediatric doctor (ST3+) attends**

- If ongoing clinical concerns call for further senior support from Paediatrics and/or ICU Consultant. Involve WATCh (0300 0300 789) early if appears very unwell or poor response to treatment and escalate to Paediatric/ICU Consultant if not already present.

6 **Consider inotropic support**

- If normal physiology is not restored after ≥20mls/kg fluid (10ml/kg in neonates)
- ICU admission
- Adrenaline infusion is usually first line *(Central strength Adrenaline & Noradrenaline infusions can be given via IO or CVL)*

If after delivering the Sepsis Six, child still has:

- Reduced consciousness despite resuscitation
- Severe tachycardia or tachypnoea
- Lactate remains > 2 mmol/l after 1hr

Or is clearly critically ill at any time

Then call for senior assistance immediately!

RDUH antibiotic guideline:

- IV Ceftriaxone or Cefotaxime (check allergies), or IV Piperacillin-Tazobactam for neutropenic sepsis
- Add in IV amoxicillin 50mg/kg in infants < 28 days
- Add in IV aciclovir if HSV is in differential
- Consider adding additional agents such as per Microguide, RDUH full sepsis guideline, UK-PAS Antimicrobial Guide and/ or microbiology advice

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